

Program: Diploma in Integrated Circuit Design and Fabrication	
Course Code: 5379B	Course Title: Analog & Digital Communication Systems Lab
Semester: 5	Credits: 1
Course Category: Program Elective	
Periods per week: 4 (L:0 T:1 P:3)	Periods per semester: 60

Course Objectives:

1. To give hands on experience and learn various electronic components, analogue modulation circuits, pulse modulation circuits.
2. To give hands on experience to implement digital modulation techniques

Course Prerequisites:

Topic	Course code	Course name	Semester
Testing of active and passive components, familiarization of equipment.	2039	Fundamentals of Electrical and Electronics Engineering lab	2
Operation of transistor, diode	2041	Basic Electronics	2
Knowledge on IC741 and other ICs	4377	Linear Integrated circuits lab	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Experiment with analog modulation schemes	15	Applying
CO2	Develop pulse modulation circuits	12	Applying
CO3	Build circuits to implement digital pulse modulator	12	Applying
CO4	Model various digital modulation techniques	15	Applying
	Lab Exam	6	Applying

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3						
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Experiment with analog modulation schemes		
M1.01	Construct an AM generator using transistor and measure the modulation index	4.5	Applying
M1.02	Construct AM demodulator using diode and observe the input-output waveforms	3	Applying
M1.03	Build an FM generator using 555 IC and observe the input-output waveforms.	3	Applying
M1.04	Build an FM generator and demodulator using 565 IC and observe the input-output waveforms	4.5	Applying
CO2	Develop Pulse modulation circuits		
M2.01	Develop a PAM modulator and demodulator circuit. Observe the input-output waveforms	3	Applying
M2.02	Develop a PWM modulator using 555 IC and observe the input-output waveforms	4.5	Applying
M2.03	Develop a PPM modulator circuit using 555 and observe the input-output waveforms	4.5	Applying
	Lab Exam 1	3	Applying
CO3	Build circuits to implement digital pulse modulator.		

M3.01	1. Construct a simple sampling circuit and observe the waveforms. 2. Reconstruct a sine wave from its samples	3	Applying
M3.02	Build a pulse code modulator circuit and observe the waveforms	6	Applying
M3.03	Develop a delta modulator and observe the input-output waveforms.	3	Applying
CO4	Model various digital modulation techniques.		
M4.01	Model a Binary Amplitude Shift Keying (modulator & demodulator) and observe the input-output waveforms.	3	Applying
M4.02	Construct a Binary Frequency Shift Keying (modulator & demodulator) and observe the input-output waveforms	4.5	Applying
M4.03	Model a Binary Phase Shift Keying (modulator & demodulator) and observe the input-output waveforms	4.5	Applying
M4.04	Construct a two channel TDM circuit (Multiplexer and Demultiplexer) and observe the waveforms	3	Applying
	Lab Exam 2	3	

Suggested Open Ended Projects

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can-do open-ended experiments as a group of 3-5. There should be no duplication in experiments between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on open ended experiment of their choice.

Example:

1. Simulation of filters using simulation software.
2. Simulate the digital modulation scheme.

Text / Reference:

T/R	Book Title/Author
T1	Electronic Communication Systems – George Kennedy – TMH
T2	Electronics Lab Manual Vol-2 KA Navas 4th edition
R1	Electronic communications - Roddy and Coolen – PHI
R2	Principles of Communication systems – Taub and Schilling

Online resources:

Sl.No.	Website Link
1	https://nptel.ac.in/courses/117/105/117105143
2	https://nptel.ac.in/courses/117101051/